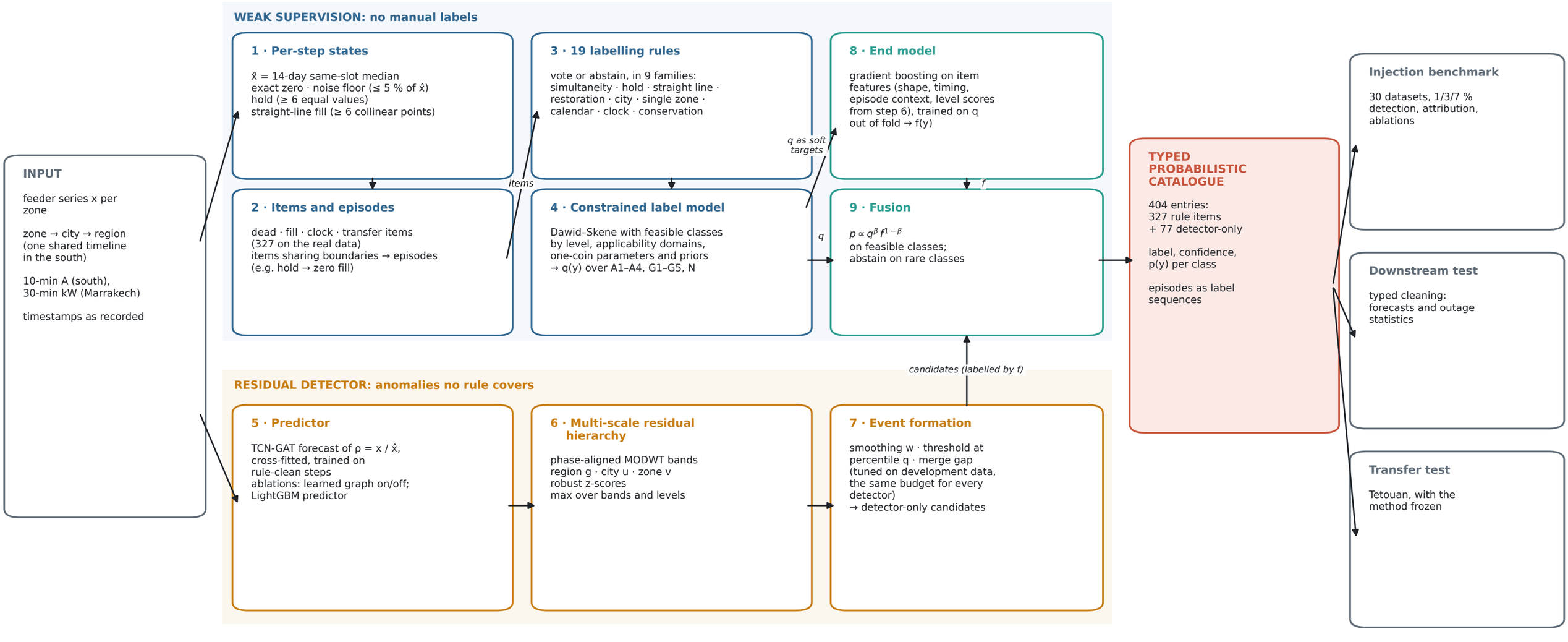
INPUT
feeder series x per zone
zone → city → region (one shared timeline in the south)
10-min A (south), 30-min kW (Marrakech)
timestamps as recorded
WEAK SUPERVISION: no manual labels
1 · Per-step states
x̂ = 14-day same-slot median
exact zero · noise floor (≤ 5 % of x̂)
hold (≥ 6 equal values)
straight-line fill (≥ 6 collinear points)
3 · 19 labelling rules
vote or abstain, in 9 families:
simultaneity · hold · straight line · restoration · city · single zone · calendar · clock · conservation
8 · End model
gradient boosting on item features (shape, timing, episode context, level scores from step 6), trained on q
out of fold → f(y)
q as soft targets
items
f
2 · Items and episodes
dead · fill · clock · transfer items (327 on the real data)
items sharing boundaries → episodes (e.g. hold → zero fill)
4 · Constrained label model
Dawid–Skene with feasible classes by level, applicability domains, one-coin parameters and priors
→ q(y) over A1–A4, G1–G5, N
q
9 · Fusion
p ∝ q^β f^(1−β)
on feasible classes; abstain on rare classes
candidates (labelled by f)
RESIDUAL DETECTOR: anomalies no rule covers
5 · Predictor
TCN-GAT forecast of ρ = x / x̂, cross-fitted, trained on rule-clean steps
ablations: learned graph on/off; LightGBM predictor
6 · Multi-scale residual hierarchy
phase-aligned MODWT bands
region g · city u · zone v
robust z-scores
max over bands and levels
7 · Event formation
smoothing w · threshold at percentile q · merge gap (tuned on development data, the same budget for every detector)
→ detector-only candidates
TYPED PROBABILISTIC CATALOGUE
404 entries:
327 rule items
+ 77 detector-only
label, confidence, p(y) per class
episodes as label sequences
Injection benchmark
30 datasets, 1/3/7 %
detection, attribution, ablations
Downstream test
typed cleaning:
forecasts and outage statistics
Transfer test
Tetouan, with the method frozen
A1 zero-fill gap · A2 simultaneous fill · A3 single-zone fill · A4 clock error · G1 regional outage · G2 city outage · G3 unplanned feeder outage · G4 planned work · G5 load transfer · N normal / untyped